Lab 4 VibeCoding - Snake (45_snake) - Chat History

AI pair-programming session with Claude Code (Claude Opus 5.5), 2026-10-23. Exported from the session transcript. Long tool inputs/outputs are truncated.

User

```
@"C:\Users\nsmr2\Downloads\Lab_4_VibeCoding_Student_handout.pdf"
https://github.com/SETAPESU26/45_snake
this is what i have to do. do everything
```

Tool call: Read

```
{
 "file_path": "C:\\Users\\nsmr2\\Downloads\\Lab_4_VibeCoding_Student_handout.pdf",
 "pages": "1-10"
}
```

```
pdftoppm is not installed. Install poppler-utils (e.g. `brew install poppler` or `apt-get install poppler-
utils`) to enable PDF page rendering.
```

Tool call: Bash

```
# Check available PDF and dev tools
python -c "import pypdf; print(pypdf.__version__)" 2>&1; python -c "import fitz; print('fitz ok')" 2>&1; which
pdftotext git gh python node 2>&1
```

```
Exit code 1
6.20.0
warning: The `fitz` API is deprecated and will be removed in future. Use `import pymupdf` instead.
fitz ok
which: no gh in (/c/Users/nsmr2/bin:/mingw64/bin:/usr/local/bin:/usr/bin:/bin:/mingw64/bin:/usr/bin:/c/
Users/nsmr2/bin:/c/Program Files/Eclipse Adoptium/jdk-8.0 ...
/mingw64/bin/pdftotext
/mingw64/bin/git
/c/Users/nsmr2/AppData/Local/Programs/Python/Python311/python
/c/Program Files/nodejs/node
```

Tool call: Bash

```
# Extract text from the lab handout PDF
pdftotext -layout "/c/Users/nsmr2/Downloads/Lab_4_VibeCoding_Student_handout.pdf" -
```

```
Lab 4: VibeCoding

Objective

Use Vibe Coding tools to fix the broken code and add features to the same. Learn how
to prompt to get the job done in under three to four attempts. Clone a repo assigned
to you, make changes and push to your repo in your personal account.

THIS IS AN INDIVIDUAL ASSIGNMENT. WORK INDIVIDUALLY!!

Duration

60 minutes

Software Requirements

     Access to Vibe Coding tool. (ChatGPT, Gemini, Claude etc)

Instructions:

... (21 more lines)
```

Tool call: Bash

```
# List working directory and git identity
cd "/c/D/B.Tech/3rd year/SE" && ls -la && git config --global user.name; git config --global user.email
```

```
total 2972
drwxr-xr-x 1 nsmr2 197609       0 Oct 20 10:27 .
drwxr-xr-x 1 nsmr2 197609       0 Oct 22 13:53 ..
-rw-r--r-- 1 nsmr2 197609  125017 Aug 31 09:54 PES1UG24AM170_Lab1_Report.docx
-rw-r--r-- 1 nsmr2 197609  402632 Aug 31 09:54 PES1UG24AM170_Lab1_Report.pdf
-rw-r--r-- 1 nsmr2 197609  356028 Oct  4 20:56 PES1UG24AM170_Lab2.docx
-rw-r--r-- 1 nsmr2 197609  270588 Oct  4 21:01 PES1UG24AM170_Lab2_Report.pdf
-rw-r--r-- 1 nsmr2 197609   30466 Oct 14 23:25 PES1UG24AM170_Lab3.docx
-rw-r--r-- 1 nsmr2 197609   81348 Oct 14 23:25 PES1UG24AM170_Lab3_Report.pdf
-rw-r--r-- 1 nsmr2 197609 1277109 Oct 14 23:09 PES1UG24AM170_Lab3_UML_Diagram.png
-rw-r--r-- 1 nsmr2 197609   16097 Aug 31 08:57 Requirements.docx
-rw-r--r-- 1 nsmr2 197609  176737 Aug 31 08:58 Requirements.pdf
-rw-r--r-- 1 nsmr2 197609   31109 Aug 31 08:52 USE_CASE_DIAGRAM.pdf
-rw-r--r-- 1 nsmr2 197609  117798 Aug 31 08:52 Use_Case_Flow.pdf
-rw-r--r-- 1 nsmr2 197609     162 Aug 31 09:28 ~$E_CASE_DIAGRAM.pdf
-rw-r--r-- 1 nsmr2 197609     162 Aug 31 09:47 ~$S1UG24AM170_Lab1_Report.docx
```

```
-rw-r--r-- 1 nsmr2 197609     162 Aug 31 09:32 ~$e_Case_Flow.pdf
-rw-r--r-- 1 nsmr2 197609     162 Aug 31 08:57 ~$quirements.docx
-rw-r--r-- 1 nsmr2 197609  124769 Aug 31 09:47 ~WRL4051.tmp
Mukesh
... (1 more lines)
```

Tool call: Bash

```
# Clone the assigned snake repo and list contents
cd "/c/D/B.Tech/3rd year/SE" && git clone https://github.com/SETAPESU26/45_snake.git && cd 45_snake && git log
--oneline -20 && ls -la && find . -path ./.git -p ...
```

```
Cloning into '45_snake'...
114601f Initial Snake  starter
total 21
drwxr-xr-x 1 nsmr2 197609    0 Oct 23 18:28 .
drwxr-xr-x 1 nsmr2 197609    0 Oct 23 18:28 ..
drwxr-xr-x 1 nsmr2 197609    0 Oct 23 18:28 .git
-rw-r--r-- 1 nsmr2 197609 2751 Oct 23 18:28 README.md
drwxr-xr-x 1 nsmr2 197609    0 Oct 23 18:28 game
-rw-r--r-- 1 nsmr2 197609  930 Oct 23 18:28 main.py
-rw-r--r-- 1 nsmr2 197609    8 Oct 23 18:28 requirements.txt
./game/food.py
./game/game_engine.py
./game/snake.py
./game/__pycache__/food.cpython-312.pyc
./game/__pycache__/game_engine.cpython-312.pyc
./game/__pycache__/snake.cpython-312.pyc
./main.py
./README.md
./requirements.txt
```

Claude

Cloned the repo — reading the README and code now.

Tool call: Bash

```
# Print README and all source files
cd "/c/D/B.Tech/3rd year/SE/45_snake" && for f in README.md requirements.txt main.py game/*.py; do echo "=====
$f"; cat -n "$f"; done
```

```
===== README.md
     1 # Snake Game
     2
     3 This project is a terminal-based Snake game using **Pygame**. It introduces students to interactive
game design using object-oriented principles and real ...
     4
     5 ---
     6
     7 ## What's Provided
     8
     9 A partially working version of a snake game with:
    10
    11 - Grid-based snake movement controlled by the player
    12 - Food spawning and snake growth
    13 - Score display
    14
    15 You are expected to **analyze**, **interact with an AI assistant**, and **complete/fix** the game to
make it fully functional.
    16
    17 ### **Use an LLM (e.g. ChatGPT or Claude) as your debugging and pair-programming partner for this
lab.**
    18
    19 ---
... (284 more lines)
```

Tool call: Bash

```
# Check Python packages for game, video, and docs
cd "/c/D/B.Tech/3rd year/SE/45_snake" && python --version && python -c "import pygame; print('pygame',
pygame.version.ver)" 2>&1 | tail -1; for m in numpy cv2 i ...
```

```
Exit code 127
Python 3.11.9
ModuleNotFoundError: No module named 'pygame'
numpy ok 2.4.6
cv2 ok 5.0.0
imageio ok 2.38.1
ModuleNotFoundError: No module named 'imageio_ffmpeg'
docx ok 1.2.0
ModuleNotFoundError: No module named 'reportlab'
ModuleNotFoundError: No module named 'markdown'
which: no ffmpeg in (/c/Users/nsmr2/bin:/mingw64/bin:/usr/local/bin:/usr/bin:/bin:/mingw64/bin:/usr/bin:/c/
Users/nsmr2/bin:/c/Program Files/Eclipse Adoptium/jdk ...
/usr/bin/bash: line 1: py: command not found
```

Claude

Installing pygame (the repo's only requirement) plus imageio-ffmpeg for recording the demo videos.

Tool call: Bash

```
# Install pygame and bundled ffmpeg
python -m pip install pygame imageio-ffmpeg 2>&1 | tail -3 && python -c "import pygame, imageio_ffmpeg;
print(pygame.version.ver, imageio_ffmpeg.get_ffmpeg_exe( ...
```

```
[notice] A new release of pip is available: 24.0 -> 26.2.2
[notice] To update, run: python.exe -m pip install --upgrade pip
pygame 2.6.1 (SDL 2.28.4, Python 3.11.9)
Hello from the pygame community. https://www.pygame.org/contribute.html
2.6.1 C:\Users\nsmr2\AppData\Local\Programs\Python\Python311\Lib\site-
packages\imageio_ffmpeg\binaries\ffmpeg-win-x86_64-v7.1.exe
```

Claude

Repo has a README with 4 tasks (collision fix, game-over screen, replay with difficulty, sound). Next I'll build a headless recorder so I can capture the "before" video from the untouched code.

Tool call: Write

```
file: C:\Users\nsmr2\AppData\Local\Temp\claude\C--Users-nsmr2--claude-spares-spare-N7tt8u\cc0425f0-5907-4bc4-
b41f-f529304b67cc\scratchpad\record_demo.py

"""Headless gameplay recorder for the Snake lab.

Runs the repo's real main.py with a dummy SDL display, drives it with scripted
key presses (posted through the normal pygame event queue), grabs every frame
from pygame.display.flip() and writes an MP4. A caption bar under the game
explains what is being shown. If the game plays sounds via game.sound, the
same samples are mixed into an audio track at the frames they were played.

usage: python record_demo.py <repo_dir> <out.mp4> <before|after> [seed]
"""
import os
import sys
import random
import subprocess
import importlib.util
import wave

os.environ["SDL_VIDEODRIVER"] = "dummy"
os.environ["SDL_AUDIODRIVER"] = "dummy"

import numpy as np
import pygame
import imageio_ffmpeg

repo, out_path, scenario_name = sys.argv[1:4]
seed = int(sys.argv[4]) if len(sys.argv) > 4 else 7
repo = os.path.abspath(repo)
sys.path.insert(0, repo)
random.seed(seed)

FPS = 60
CAPTION_H = 72

KEY_NAMES = {
    pygame.K_UP: "UP", pygame.K_DOWN: "DOWN", pygame.K_LEFT: "LEFT", pygame.K_RIGHT: "RIGHT",
    pygame.K_RETURN: "ENTER", pygame.K_ESCAPE: "ESC",
    pygame.K_1: "1", pygame.K_2: "2", pygame.K_3: "3", pygame.K_q: "Q",
}
... (327 more lines)
```

```
File created successfully at: C:\Users\nsmr2\AppData\Local\Temp\claude\C--Users-nsmr2--claude-spares-spare-
N7tt8u\cc0425f0-5907-4bc4-b41f-f529304b67cc\scratchpa ...
```

Tool call: Bash

```
# Record the 'before' gameplay video from original code
cd "/c/D/B.Tech/3rd year/SE" && mkdir -p Lab-4/videos && SP="/c/Users/nsmr2/AppData/Local/Temp/claude/C--
Users-nsmr2--claude-spares-spare-N7tt8u/cc0425f0-5907-4 ...
```

```
Game over! Final score: 2
wrote Lab-4/videos/before.mp4: 601 frames (10.0s), marks={'rev': 370, 'dead': 377}, sounds=[]
```

Tool call: Bash

```
# Extract preview frames from before video
cd "/c/D/B.Tech/3rd year/SE" && SP="/c/Users/nsmr2/AppData/Local/Temp/claude/C--Users-nsmr2--claude-spares-
spare-N7tt8u/cc0425f0-5907-4bc4-b41f-f529304b67cc/scr ...
```

```
ok
```

Tool call: Read

```
{
 "file_path": "C:\\Users\\nsmr2\\AppData\\Local\\Temp\\claude\\C--Users-nsmr2--claude-spares-spare-N7tt8u\
\cc0425f0-5907-4bc4-b41f-f529304b67cc\\scratchpad\\bef ...
```

```
}
```

[image]

Tool call: Edit

```
file: C:\Users\nsmr2\AppData\Local\Temp\claude\C--Users-nsmr2--claude-spares-spare-N7tt8u\cc0425f0-5907-4bc4-
b41f-f529304b67cc\scratchpad\record_demo.py
--- old
        self.real = real
        self.last_line = ""

    def write(self, s):
        self.real.write(s)
        for line in s.splitlines():
            if line.strip():
                self.last_line = line.strip()
+++ new
        self.real = real
        self.last_line = ""
        self._buf = ""

    def write(self, s):
        self.real.write(s)
        self._buf += s
        while "\n" in self._buf:
            line, self._buf = self._buf.split("\n", 1)
            if line.strip():
                self.last_line = line.strip()
```

```
The file C:\Users\nsmr2\AppData\Local\Temp\claude\C--Users-nsmr2--claude-spares-spare-
N7tt8u\cc0425f0-5907-4bc4-b41f-f529304b67cc\scratchpad\record_demo.py has  ...
```

Tool call: Bash

```
# Re-record before video and extract preview frames
cd "/c/D/B.Tech/3rd year/SE" && SP="/c/Users/nsmr2/AppData/Local/Temp/claude/C--Users-nsmr2--claude-spares-
spare-N7tt8u/cc0425f0-5907-4bc4-b41f-f529304b67cc/scr ...
```

```
Game over! Final score: 2
wrote Lab-4/videos/before.mp4: 601 frames (10.0s), marks={'rev': 370, 'dead': 377}, sounds=[]
ok
```

Claude

Before video recorded (10 s, shows the reverse-key instant death and the frozen no-game-over state). Checking the frames, then moving on to the fixes.

Tool call: Read

```
{
 "file_path": "C:\\Users\\nsmr2\\AppData\\Local\\Temp\\claude\\C--Users-nsmr2--claude-spares-spare-N7tt8u\
\cc0425f0-5907-4bc4-b41f-f529304b67cc\\scratchpad\\bef ...
}
```

[image]

Tool call: Bash

```
# Rename origin and disable pushing to the class repo
cd "/c/D/B.Tech/3rd year/SE/45_snake" && git remote rename origin upstream && git remote set-url --push
upstream DISABLED-do-not-push-to-SETAPESU26 && git remot ...
```

```
upstream https://github.com/SETAPESU26/45_snake.git (fetch)
upstream DISABLED-do-not-push-to-SETAPESU26 (push)
```

Tool call: Write

```
file: C:\D\B.Tech\3rd year\SE\45_snake\game\snake.py

import pygame

# Turns queued between two moves; enough for a quick "up then left" double tap.
MAX_QUEUED_TURNS = 2

class Snake:
    def __init__(self, x, y, cell_size):
        self.cell_size = cell_size
        # body is a list of (x, y) grid-cell positions, head is body[0]
        self.body = [(x, y), (x - 1, y), (x - 2, y)]
        self.direction = (1, 0)  # direction of the last move (moving right)
        self.pending_turns = []  # key presses not yet applied to a move
        self.grow_pending = False

    def set_direction(self, dx, dy):
        # Validate against the last queued turn (or the current heading), not
        # just the current heading: otherwise two quick presses between moves
        # (e.g. Up then Left while moving Right) still reverse into the body.
        last = self.pending_turns[-1] if self.pending_turns else self.direction
        if (dx, dy) == last or (dx, dy) == (-last[0], -last[1]):
            return
```

```
        if len(self.pending_turns) < MAX_QUEUED_TURNS:
            self.pending_turns.append((dx, dy))

    def next_head(self):
        # Where the head will be after the next move, without moving.
        dx, dy = self.pending_turns[0] if self.pending_turns else self.direction
        head_x, head_y = self.body[0]
        return (head_x + dx, head_y + dy)

    def move(self):
        new_head = self.next_head()
        if self.pending_turns:
            self.direction = self.pending_turns.pop(0)

        self.body.insert(0, new_head)
        if self.grow_pending:
            self.grow_pending = False
... (25 more lines)
```

The file C:\D\B.Tech\3rd year\SE\45_snake\game\snake.py has been updated successfully. (file state is current in your context — no need to Read it back)

Tool call: Edit

```
file: C:\D\B.Tech\3rd year\SE\45_snake\game\game_engine.py
--- old
        self._frame_counter = 0

        self.snake.move()

        if self.snake.collides_with_wall(self.grid_width, self.grid_height):
            self.game_over = True
            return

        if self.snake.collides_with_self():
            self.game_over = True
            return

        if self.snake.head_rect().colliderect(self.food.rect()):
            self.snake.grow()
+++ new
        self._frame_counter = 0

        # Check the cell the head is about to enter before moving, so the
        # snake stays on screen where it crashed instead of off the grid.
        next_head = self.snake.next_head()
        if self.snake.collides_with_wall(next_head, self.grid_width, self.grid_height):
            self.game_over = True
            return

        if self.snake.collides_with_self(next_head):
            self.game_over = True
            return

        self.snake.move()

        if self.snake.body[0] == (self.food.x, self.food.y):
            self.snake.grow()
```

Tool call: Bash

```
# Show the keydown handler section
cd "/c/D/B.Tech/3rd year/SE/45_snake" && sed -n 31,46p game/game_engine.py
```

The file C:\D\B.Tech\3rd year\SE\45_snake\game\game_engine.py has been updated successfully.

```
    def handle_keydown(self, key):
            # Direction changes are applied immediately on key press.
            if key in (pygame.K_UP, pygame.K_w):
                self.snake.set_direction(0, -1)
            elif key in (pygame.K_DOWN, pygame.K_s):
                self.snake.set_direction(0, 1)
            elif key in (pygame.K_LEFT, pygame.K_a):
                self.snake.set_direction(-1, 0)
            elif key in (pygame.K_RIGHT, pygame.K_d):
                self.snake.set_direction(1, 0)

        def handle_input(self):
            # Reserved for continuously-held-key input (not used for a
            # grid-based snake, but kept here to mirror the engine's shape).
            pass
```

Tool call: Edit

```
file: C:\D\B.Tech\3rd year\SE\45_snake\game\game_engine.py
--- old
        # Direction changes are applied immediately on key press.
+++ new
```

```
        # Direction changes are queued and applied on the next move(s);
        # the snake rejects reversals into its own body.
```

```
The file C:\D\B.Tech\3rd year\SE\45_snake\game\game_engine.py has been updated successfully.
```

Tool call: Write

```
file: C:\Users\nsmr2\AppData\Local\Temp\claude\C--Users-nsmr2--claude-spares-spare-N7tt8u\cc0425f0-5907-4bc4-
b41f-f529304b67cc\scratchpad\test_task1.py

import os
import sys

os.environ["SDL_VIDEODRIVER"] = "dummy"
os.environ["SDL_AUDIODRIVER"] = "dummy"
sys.path.insert(0, sys.argv[1])

import pygame

pygame.init()
from game.snake import Snake
from game.game_engine import GameEngine


def tick(engine):
    # advance exactly one snake move
    engine._frame_counter = 10_000
    engine.update()


# 1. reversal is ignored
s = Snake(10, 10, 20)
s.set_direction(-1, 0)
s.move()
assert s.body[0] == (11, 10), s.body

# 2. quick Up then Left between moves -> two turns, no reversal
s = Snake(10, 10, 20)
s.set_direction(0, -1)
s.set_direction(-1, 0)
s.move()
assert s.body[0] == (10, 9), s.body
s.move()
assert s.body[0] == (9, 9), s.body
assert not s.collides_with_self(s.next_head())

# 3. Up then Down quickly -> Down rejected (opposite of queued Up)
s = Snake(10, 10, 20)
... (38 more lines)
```

```
File created successfully at: C:\Users\nsmr2\AppData\Local\Temp\claude\C--Users-nsmr2--claude-spares-spare-
N7tt8u\cc0425f0-5907-4bc4-b41f-f529304b67cc\scratchpa ...
```

Tool call: Bash

```
# Run Task 1 collision tests
cd "/c/D/B.Tech/3rd year/SE/45_snake" && python -B "/c/Users/nsmr2/AppData/Local/Temp/claude/C--Users-nsmr2--
claude-spares-spare-N7tt8u/cc0425f0-5907-4bc4-b41f- ...
```

```
task1 tests passed
```

Claude

Task 1 fix passes my tests (reverse key ignored, quick double-taps buffered, tail-following allowed, snake stops on-grid at walls). Committing it.

Tool call: Bash

```
# Check working tree before committing
cd "/c/D/B.Tech/3rd year/SE/45_snake" && git status --short && ls game/__pycache__ && git diff --stat
```

```
M game/game_engine.py
 M game/snake.py
food.cpython-312.pyc
game_engine.cpython-312.pyc
snake.cpython-312.pyc
warning: in the working copy of 'game/snake.py', LF will be replaced by CRLF the next time Git touches it
 game/game_engine.py | 16 ++++++++++------
 game/snake.py       | 41 +++++++++++++++++++++++++++++++++++------------
 2 files changed, 39 insertions(+), 18 deletions(-)
```

Tool call: Bash

```
# Commit Task 1 collision fixes
cd "/c/D/B.Tech/3rd year/SE/45_snake" && git add game/snake.py game/game_engine.py && git commit -q -F -
<<'EOF'
Task 1: Refine collision detection

- Ignore direction changes that reverse straight into the body.
```

```
- Validate turns against the last queued turn, not just the current
  heading, so two quick presses between moves (e.g. Up then Left while
  moving Right) can no longer cause a reversal; queue up to two turns so
  quick double taps are not lost.
- Check wall/self collisions on the cell the head is about to enter,
  before moving, so the snake stays visible where it crashed.
- Treat the cell the tail is vacating as free unless the snake is
  growing, so following your own tail is not an unfair death.
- Detect eating by grid cell instead of rectangle overlap.

Co-Authored-By: Claude Opus 5.5 <noreply@anthropic.com>
EOF
git log --oneline -3
```

```
warning: in the working copy of 'game/snake.py', LF will be replaced by CRLF the next time Git touches it
ceb9fc9 Task 1: Refine collision detection
114601f Initial Snake  starter
```

Tool call: Write

```
file: C:\D\B.Tech\3rd year\SE\45_snake\game\food.py

import pygame
import random

class Food:
    def __init__(self, grid_width, grid_height, cell_size, occupied_cells=()):
        self.grid_width = grid_width
        self.grid_height = grid_height
        self.cell_size = cell_size
        self.x = 0
        self.y = 0
        self.respawn(occupied_cells)

    def respawn(self, occupied_cells):
        # Pick from the free cells directly: never lands on the snake, and
        # can't loop forever when the board is (almost) full.
        occupied = set(occupied_cells)
        free_cells = [
            (x, y)
            for x in range(self.grid_width)
            for y in range(self.grid_height)
            if (x, y) not in occupied
        ]
        if not free_cells:
            return False
        self.x, self.y = random.choice(free_cells)
        return True

    def rect(self):
        return pygame.Rect(self.x * self.cell_size, self.y * self.cell_size, self.cell_size, self.cell_size)
```

Tool call: Edit

```
file: C:\D\B.Tech\3rd year\SE\45_snake\game\game_engine.py
--- old
        self.food = Food(self.grid_width, self.grid_height, self.cell_size)
+++ new
        self.food = Food(self.grid_width, self.grid_height, self.cell_size, self.snake.body)
```

```
The file C:\D\B.Tech\3rd year\SE\45_snake\game\food.py has been updated successfully. (file state is
current in your context — no need to Read it back)
```

```
The file C:\D\B.Tech\3rd year\SE\45_snake\game\game_engine.py has been updated successfully.
```

Tool call: Bash

```
# Find food respawn call in engine
cd "/c/D/B.Tech/3rd year/SE/45_snake" && grep -n "respawn" game/game_engine.py
```

```
74:            self.food.respawn(self.snake.body)
```

Tool call: Edit

```
file: C:\D\B.Tech\3rd year\SE\45_snake\game\game_engine.py
--- old
            self.food.respawn(self.snake.body)
+++ new
            if not self.food.respawn(self.snake.body):
                # Snake fills the whole board: nothing left to eat.
                self.game_over = True
```

Tool call: Bash

```
# Test food spawning never lands on the snake
cd "/c/D/B.Tech/3rd year/SE/45_snake" && cat > "/c/Users/nsmr2/AppData/Local/Temp/claude/C--Users-nsmr2--
claude-spares-spare-N7tt8u/cc0425f0-5907-4bc4-b41f-f529 ...
import os, sys, random
os.environ["SDL_VIDEODRIVER"] = "dummy"; os.environ["SDL_AUDIODRIVER"] = "dummy"
sys.path.insert(0, sys.argv[1])
```

```
import pygame; pygame.init()
from game.food import Food
from game.game_engine import GameEngine
for seed in range(3000):
    random.seed(seed)
    e = GameEngine(600, 600)
    assert (e.food.x, e.food.y) not in e.snake.body, seed
f = Food(2, 2, 20, [(0, 0), (0, 1), (1, 0)])
assert (f.x, f.y) == (1, 1)
assert f.respawn([(0, 0), (0, 1), (1, 0), (1, 1)]) is False
print("food tests passed")
EOF
python -B "/c/Users/nsmr2/AppData/Local/Temp/claude/C--Users-nsmr2--claude-spares-spare-N7tt8u/
cc0425f0-5907-4bc4-b41f-f529304b67cc/scratchpad/test_food.py" . 2 ...
```

```
The file C:\D\B.Tech\3rd year\SE\45_snake\game\game_engine.py has been updated successfully.
```

```
food tests passed
task1 tests passed
```

Tool call: Bash

```
# Commit food spawn fix
cd "/c/D/B.Tech/3rd year/SE/45_snake" && git add game/food.py game/game_engine.py && git commit -q -F -
<<'EOF'
Fix food spawning on top of the snake

The first food was spawned with an empty occupied list, so it could
appear on the starting snake. Pass the snake body in, and pick from the
free cells directly so respawn never loops forever on a full board.

Co-Authored-By: Claude Opus 5.5 <noreply@anthropic.com>
EOF
git log --oneline -3
```

```
warning: in the working copy of 'game/food.py', LF will be replaced by CRLF the next time Git touches it
3a63280 Fix food spawning on top of the snake
ceb9fc9 Task 1: Refine collision detection
114601f Initial Snake  starter
```

Claude

Task 1 and a food-spawn fix are committed. Now building the Task 2 game-over screen.

Tool call: Read

```
{
 "file_path": "C:\\D\\B.Tech\\3rd year\\SE\\45_snake\\game\\game_engine.py"
}
```

```
1 import pygame
2 from .snake import Snake
3 from .food import Food
4
5 # Game Engine
6
7 WHITE = (255, 255, 255)
8 GREEN = (0, 200, 0)
9 RED = (220, 60, 60)
10
11 class GameEngine:
12     def __init__(self, width, height):
13         self.width = width
14         self.height = height
15         self.cell_size = 20
16         self.grid_width = width // self.cell_size
17         self.grid_height = height // self.cell_size
18
19         self.snake = Snake(self.grid_width // 2, self.grid_height // 2, self.cell_size)
20         self.food = Food(self.grid_width, self.grid_height, self.cell_size, self.snake.body)
... (74 more lines)
```

Tool call: Bash

```
# Add game-over screen to the engine
cd "/c/D/B.Tech/3rd year/SE/45_snake" && python - <<'EOF'
from pathlib import Path
p = Path("game/game_engine.py")
s = p.read_text()

def sub(old, new):
    global s
    assert s.count(old) == 1, old
    s = s.replace(old, new)

sub('''RED = (220, 60, 60)
''', '''RED = (220, 60, 60)
GRAY = (170, 170, 170)
OVERLAY = (0, 0, 0, 180)
```

```
''')
sub('''        self.font = pygame.font.SysFont("Arial", 30)
''', '''        self.font = pygame.font.SysFont("Arial", 30)
        self.title_font = pygame.font.SysFont("Arial", 64, bold=True)
        self.small_font = pygame.font.SysFont("Arial", 24)
''')
sub('''        self.game_over = False
        self._game_over_logged = False

    def handle_keydown(self, key):
''', '''        self.game_over = False

    def handle_keydown(self, key):
        if self.game_over:
            # The game-over screen waits here until the player decides.
            if key in (pygame.K_ESCAPE, pygame.K_q):
                pygame.event.post(pygame.event.Event(pygame.QUIT))
            return

''')
sub('''        if self.snake.collides_with_wall(next_head, self.grid_width, self.grid_height):
            self.game_over = True
            return

        if self.snake.collides_with_self(next_head):
... (43 more lines)
```

```
 game/game_engine.py | 41 ++++++++++++++++++++++++++++++++++++++--------
 1 file changed, 33 insertions(+), 8 deletions(-)
```

Tool call: Bash

```
# Test game-over screen and render a screenshot
cd "/c/D/B.Tech/3rd year/SE/45_snake" && SP="/c/Users/nsmr2/AppData/Local/Temp/claude/C--Users-nsmr2--claude-
spares-spare-N7tt8u/cc0425f0-5907-4bc4-b41f-f529304 ...
import os, sys
os.environ["SDL_VIDEODRIVER"] = "dummy"; os.environ["SDL_AUDIODRIVER"] = "dummy"
sys.path.insert(0, sys.argv[1])
import pygame; pygame.init()
screen = pygame.display.set_mode((600, 600))
from game.game_engine import GameEngine
e = GameEngine(600, 600)
e.score = 7
e.snake.body = [(29, 15), (28, 15), (27, 15), (26, 15)]
e._frame_counter = 999
e.update()
assert e.game_over
# keys while game over don't crash / don't steer; Esc posts QUIT
pygame.event.clear()
e.handle_keydown(pygame.K_LEFT)
assert not pygame.event.get(pygame.QUIT)
e.handle_keydown(pygame.K_ESCAPE)
assert pygame.event.get(pygame.QUIT)
screen.fill((0, 0, 0))
e.render(screen)
pygame.image.save(screen, sys.argv[2])
print("task2 tests passed")
EOF
python -B "$SP/test_task2.py" . "$SP/gameover_t2.png" 2>&1 | grep -v "Hello from\|pygame 2.6" && python -B
"$SP/test_task1.py" . 2>&1 | grep -v "Hello from\|pyg ...
```

```
 task2 tests passed
 task1 tests passed
```

Tool call: Read

```
{
 "file_path": "C:\\Users\\nsmr2\\AppData\\Local\\Temp\\claude\\C--Users-nsmr2--claude-spares-spare-N7tt8u\
\cc0425f0-5907-4bc4-b41f-f529304b67cc\\scratchpad\\gam ...
}
```

```
 [image]
```

Tool call: Bash

```
# Commit Task 2 game-over screen
cd "/c/D/B.Tech/3rd year/SE/45_snake" && git add game/game_engine.py && git commit -q -F - <<'EOF'
Task 2: Implement Game Over screen

Replace the console print with an in-game Game Over screen: the final
board is dimmed and "GAME OVER" plus the final score are drawn on top.
The game then waits for input instead of continuing silently; steering
keys are ignored and Esc/Q (or closing the window) quits cleanly.

Co-Authored-By: Claude Opus 5.5 <noreply@anthropic.com>
EOF
git log --oneline -1
```

```
 bfb15f5 Task 2: Implement Game Over screen
```

Claude

Task 2 committed. Now Task 3: difficulty menu on the game-over screen to replay or exit.

Tool call: Read

```
{
 "file_path": "C:\\D\\B.Tech\\3rd year\\SE\\45_snake\\game\\game_engine.py",
 "limit": 60
}
```

```
   1 import pygame
   2 from .snake import Snake
   3 from .food import Food
   4
   5 # Game Engine
   6
   7 WHITE = (255, 255, 255)
   8 GREEN = (0, 200, 0)
   9 RED = (220, 60, 60)
   10 GRAY = (170, 170, 170)
   11 OVERLAY = (0, 0, 0, 180)
   12
   13 class GameEngine:
   14     def __init__(self, width, height):
   15         self.width = width
   16         self.height = height
   17         self.cell_size = 20
   18         self.grid_width = width // self.cell_size
   19         self.grid_height = height // self.cell_size
   20
   ... (40 more lines)
```

Tool call: Bash

```
# Add replay difficulty menu to the engine
cd "/c/D/B.Tech/3rd year/SE/45_snake" && python - <<'EOF'
from pathlib import Path
p = Path("game/game_engine.py")
s = p.read_text()

def sub(old, new):
    global s
    assert s.count(old) == 1, old
    s = s.replace(old, new)

sub('''GRAY = (170, 170, 170)
OVERLAY = (0, 0, 0, 180)
''', '''GRAY = (170, 170, 170)
YELLOW = (255, 220, 90)
OVERLAY = (0, 0, 0, 180)

# (name, snake moves per second); Medium is the original speed.
DIFFICULTIES = [("Easy", 5), ("Medium", 8), ("Hard", 12)]
DEFAULT_DIFFICULTY = 1
# The game-over menu lists every difficulty followed by an Exit entry.
EXIT_OPTION = len(DIFFICULTIES)
''')
sub('''        self.grid_height = height // self.cell_size

        self.snake = Snake(self.grid_width // 2, self.grid_height // 2, self.cell_size)
        self.food = Food(self.grid_width, self.grid_height, self.cell_size, self.snake.body)

        self.score = 0
        self.font = pygame.font.SysFont("Arial", 30)
        self.title_font = pygame.font.SysFont("Arial", 64, bold=True)
        self.small_font = pygame.font.SysFont("Arial", 24)

        self.moves_per_second = 8
        self._frame_counter = 0

        self.game_over = False

    def handle_keydown(self, key):
        if self.game_over:
... (86 more lines)
```

```
 game/game_engine.py | 73 +++++++++++++++++++++++++++++++++++++++++++++---------
  1 file changed, 61 insertions(+), 12 deletions(-)
```

Tool call: Bash

```
# Run Task 1-3 tests and render the menu
cd "/c/D/B.Tech/3rd year/SE/45_snake" && SP="/c/Users/nsmr2/AppData/Local/Temp/claude/C--Users-nsmr2--claude-
spares-spare-N7tt8u/cc0425f0-5907-4bc4-b41f-f529304 ...
import os, sys
os.environ["SDL_VIDEODRIVER"] = "dummy"; os.environ["SDL_AUDIODRIVER"] = "dummy"
sys.path.insert(0, sys.argv[1])
```

```
import pygame; pygame.init()
screen = pygame.display.set_mode((600, 600))
from game.game_engine import GameEngine

def crash(e):
    e.snake.body = [(29, 15), (28, 15), (27, 15)]
    e._frame_counter = 999
    e.update()
    assert e.game_over

e = GameEngine(600, 600)
assert e.moves_per_second == 8 and e.difficulty == 1
e.score = 5
crash(e)
assert e.menu_index == 1
# Down + Enter -> Hard, fresh round
e.handle_keydown(pygame.K_DOWN)
assert e.menu_index == 2
screen.fill((0, 0, 0)); e.render(screen); pygame.image.save(screen, sys.argv[2])
e.handle_keydown(pygame.K_RETURN)
assert not e.game_over and e.score == 0 and e.moves_per_second == 12 and len(e.snake.body) == 3
assert (e.food.x, e.food.y) not in e.snake.body
# number key shortcut -> Easy
crash(e)
e.handle_keydown(pygame.K_1)
assert not e.game_over and e.moves_per_second == 5
# menu wraps; Exit option posts QUIT
crash(e)
pygame.event.clear()
e.handle_keydown(pygame.K_UP)  # 0 -> 3 (Exit)
assert e.menu_index == 3
e.handle_keydown(pygame.K_RETURN)
assert pygame.event.get(pygame.QUIT)
pygame.event.clear()
e.handle_keydown(pygame.K_q)
... (13 more lines)
```

```
  task1 tests passed
  task2 tests passed
  task3 tests passed
```

Tool call: Read

```
{
 "file_path": "C:\\Users\\nsmr2\\AppData\\Local\\Temp\\claude\\C--Users-nsmr2--claude-spares-spare-N7tt8u\
\cc0425f0-5907-4bc4-b41f-f529304b67cc\\scratchpad\\men ...
}
```

```
  [image]
```

Tool call: Bash

```
# Commit Task 3 replay menu
cd "/c/D/B.Tech/3rd year/SE/45_snake" && git add game/game_engine.py && git commit -q -F - <<'EOF'
Task 3: Add replay option with difficulty selection

The Game Over screen now offers Easy (5 moves/s), Medium (8, the
original speed) or Hard (12) to play again, or Exit. Choose with
Up/Down (or W/S) + Enter, or press 1/2/3 directly; Q/Esc exits. A new
reset() starts a fresh round (snake, food, score, speed) without
restarting the program, and the current difficulty is shown in the
top-right corner during play.

Co-Authored-By: Claude Opus 5.5 <noreply@anthropic.com>
EOF
git log --oneline -1
```

```
  af8034d Task 3: Add replay option with difficulty selection
```

Claude

Task 3 committed. Now Task 4: sound effects, generated in code so no audio asset files are needed.

Tool call: Write

```
file: C:\D\B.Tech\3rd year\SE\45_snake\game\sound.py

import io
import math
import struct
import wave

import pygame

# Sound effects are synthesised at start-up, so no audio files are needed.

SAMPLE_RATE = 22050
```

```
# Each effect is a list of (start Hz, end Hz, seconds) frequency sweeps.
EAT_NOTES = [(660, 660, 0.05), (990, 1320, 0.08)]
GAME_OVER_NOTES = [(440, 440, 0.16), (370, 370, 0.16), (311, 311, 0.16), (262, 196, 0.5)]


def _synthesize(notes, volume, square):
    samples = bytearray()
    phase = 0.0
    for start_hz, end_hz, seconds in notes:
        count = int(SAMPLE_RATE * seconds)
        for i in range(count):
            freq = start_hz + (end_hz - start_hz) * i / count
            phase += 2 * math.pi * freq / SAMPLE_RATE
            wave_value = math.sin(phase)
            if square:
                wave_value = 0.6 if wave_value >= 0 else -0.6
            # Short fade in/out on every note so they don't click.
            envelope = min(1.0, i / 150, (count - i) / 300)
            samples += struct.pack("<h", int(volume * envelope * wave_value * 32767))

    buffer = io.BytesIO()
    with wave.open(buffer, "wb") as wav:
        wav.setnchannels(1)
        wav.setsampwidth(2)
        wav.setframerate(SAMPLE_RATE)
        wav.writeframes(bytes(samples))
    buffer.seek(0)
... (20 more lines)
```

Tool call: Bash

```
# Hook sound effects into the engine
cd "/c/D/B.Tech/3rd year/SE/45_snake" && python - <<'EOF'
from pathlib import Path
p = Path("game/game_engine.py")
s = p.read_text()

def sub(old, new):
    global s
    assert s.count(old) == 1, old
    s = s.replace(old, new)

sub('''from .food import Food
''', '''from .food import Food
from .sound import SoundManager
''')
sub('''        self.small_font = pygame.font.SysFont("Arial", 24)

        self.reset(DEFAULT_DIFFICULTY)
''', '''        self.small_font = pygame.font.SysFont("Arial", 24)
        self.sounds = SoundManager()

        self.reset(DEFAULT_DIFFICULTY)
''')
sub('''            self.snake.grow()
            self.score += 1
''', '''            self.snake.grow()
            self.score += 1
            self.sounds.play("eat")
''')
sub('''    def _end_game(self):
        self.game_over = True
''', '''    def _end_game(self):
        self.game_over = True
        self.sounds.play("game_over")
''')
p.write_text(s)
EOF
git diff --stat
```

```
File created successfully at: C:\D\B.Tech\3rd year\SE\45_snake\game\sound.py (file state is current in your
context — no need to Read it back)
```

```
game/game_engine.py | 4 ++++
 1 file changed, 4 insertions(+)
```

Tool call: Bash

```
# Run all task tests including sound
cd "/c/D/B.Tech/3rd year/SE/45_snake" && SP="/c/Users/nsmr2/AppData/Local/Temp/claude/C--Users-nsmr2--claude-
spares-spare-N7tt8u/cc0425f0-5907-4bc4-b41f-f529304 ...
import os, sys, time
os.environ["SDL_VIDEODRIVER"] = "dummy"; os.environ["SDL_AUDIODRIVER"] = "dummy"
sys.path.insert(0, sys.argv[1])
import pygame; pygame.init()
print("mixer:", pygame.mixer.get_init())
import game.sound as gs
```

```
from game.game_engine import GameEngine
t = time.perf_counter()
e = GameEngine(600, 600)
print(f"engine init {1000*(time.perf_counter()-t):.0f} ms")
assert set(e.sounds.sounds) == {"eat", "game_over"}
print({k: round(v.get_length(), 2) for k, v in e.sounds.sounds.items()})
played = []
orig = gs.SoundManager.play
gs.SoundManager.play = lambda self, n: (played.append(n), orig(self, n))
e.food.x, e.food.y = e.snake.next_head(); e._frame_counter = 999; e.update()
e.snake.body = [(29, 15), (28, 15), (27, 15)]; e._frame_counter = 999; e.update()
assert played == ["eat", "game_over"], played
# no audio device -> silent, not a crash
pygame.mixer.quit()
real_init = pygame.mixer.init
def broken(*a, **k): raise pygame.error("no audio device")
pygame.mixer.init = broken
m = gs.SoundManager()
assert m.sounds == {}
m.play("eat")
pygame.mixer.init = real_init
print("task4 tests passed")
EOF
for t in task1 task2 task3 task4; do python -B "$SP/test_$t.py" . "$SP/menu_$t.png" 2>&1 | grep -v "Hello
from\|pygame 2.6"; done
```

```
task1 tests passed
task2 tests passed
task3 tests passed
mixer: (44100, -16, 2)
engine init 33 ms
{'eat': 0.13, 'game_over': 0.98}
task4 tests passed
```

Tool call: Bash

```
# Commit Task 4 sound effects
cd "/c/D/B.Tech/3rd year/SE/45_snake" && git add game/sound.py game/game_engine.py && git commit -q -F -
<<'EOF'
Task 4: Add sound feedback

Play a short rising blip when food is eaten and a descending tune at
game over. The effects are synthesised in memory at start-up (no audio
files to ship), and if no audio device is available the game simply
stays silent instead of crashing.

Co-Authored-By: Claude Opus 5.5 <noreply@anthropic.com>
EOF
git log --oneline -1
```

```
warning: in the working copy of 'game/sound.py', LF will be replaced by CRLF the next time Git touches it
42fe755 Task 4: Add sound feedback
```

Claude

All four tasks committed. Quick smoke run of the real game window (opens for ~2 s), then the "after" video.

Tool call: Bash

```
# Smoke-test the real game window and audio
cd "/c/D/B.Tech/3rd year/SE/45_snake" && python -B - <<'EOF' 2>&1 | grep -v "Hello from\|pygame 2.6"
import importlib.util, sys, pygame
sys.path.insert(0, ".")
frames = [0]
real_flip = pygame.display.flip
def flip():
    real_flip()
    frames[0] += 1
    if frames[0] == 20: pygame.event.post(pygame.event.Event(pygame.KEYDOWN, key=pygame.K_LEFT))
    if frames[0] == 120: pygame.event.post(pygame.event.Event(pygame.QUIT))
pygame.display.flip = flip
spec = importlib.util.spec_from_file_location("m", "main.py")
m = importlib.util.module_from_spec(spec); spec.loader.exec_module(m)
print("video driver:", pygame.display.get_driver(), "| mixer:", pygame.mixer.get_init(), "| sounds:",
list(m.engine.sounds.sounds))
m.engine.sounds.play("eat")
m.main()
print("ran", frames[0], "frames; game_over =", m.engine.game_over)
EOF
```

```
video driver: windows | mixer: (44100, -16, 2) | sounds: ['eat', 'game_over']
ran 121 frames; game_over = True
```

Tool call: Edit

```
file: C:\Users\nsmr2\AppData\Local\Temp\claude\C--Users-nsmr2--claude-spares-spare-N7tt8u\cc0425f0-5907-4bc4-
b41f-f529304b67cc\scratchpad\record_demo.py
--- old
```

```
        elif m == "rev_ignored":
            caption(f"Task 1: {self.rev_text}", "-> reverse is ignored, no unfair death")
            if f - self.t["rev"] >= 45 and self.moved(engine):
                self.mode = "auto2"
+++ new
        elif m == "rev_ignored":
            caption(f"Task 1: {self.rev_text}", "-> reverse is ignored, no unfair death")
            if f - self.t["rev"] >= 45 and self.moved(engine):
                d = engine.snake.direction
                hx, hy = engine.snake.body[0]
                if d[1] == 0 and 2 < hy and 2 < hx < engine.grid_width - 3:
                    perp, rev = (0, -1), (-d[0], 0)
                    post_key(DIR_KEYS[perp])
                    post_key(DIR_KEYS[rev])
                    self.tap_text = f"Quick {DIR_NAMES[perp]} + {DIR_NAMES[rev]} between two moves"
                    self.mode = "double_tap"
                    self.t["tap"] = f
                else:
                    self.mode = "auto2"
        elif m == "double_tap":
            caption(f"Task 1: {self.tap_text}", "-> both turns are buffered: a safe U-turn, not a reversal")
            if f - self.t["tap"] >= 50 and self.moved(engine):
                self.mode = "auto2"
```

```
The file C:\Users\nsmr2\AppData\Local\Temp\claude\C--Users-nsmr2--claude-spares-spare-
N7tt8u\cc0425f0-5907-4bc4-b41f-f529304b67cc\scratchpad\record_demo.py has  ...
```

Tool call: Bash

```
# Record the 'after' gameplay video
cd "/c/D/B.Tech/3rd year/SE" && SP="/c/Users/nsmr2/AppData/Local/Temp/claude/C--Users-nsmr2--claude-spares-
spare-N7tt8u/cc0425f0-5907-4bc4-b41f-f529304b67cc/scr ...
```

```
wrote Lab-4/videos/after.mp4: 601 frames (10.0s), marks={'rev': 125, 'dead': 237, 'hard': 362},
sounds=[(125, 'eat'), (202, 'eat'), (237, 'game_over'), (427, 'e ...
```

Tool call: Bash

```
# Debug why the double-tap demo did not trigger
cd "/c/D/B.Tech/3rd year/SE" && SP="/c/Users/nsmr2/AppData/Local/Temp/claude/C--Users-nsmr2--claude-spares-
spare-N7tt8u/cc0425f0-5907-4bc4-b41f-f529304b67cc/scr ...
import sys
p = sys.argv[1]
s = open(p).read()
old = '''                else:
                    self.mode = "auto2"
        elif m == "double_tap":'''
new = '''                else:
                    print("no double tap: dir", d, "head", (hx, hy))
                    self.mode = "auto2"
        elif m == "double_tap":'''
assert s.count(old) == 1
open(p, "w").write(s.replace(old, new))
EOF
python -B "$SP/record_demo.py" 45_snake "$SP/after_dbg.mp4" after 7 2>&1 | grep -v "Hello from\|pygame 2.6"
```

```
no double tap: dir (-1, 0) head (5, 1)
wrote C:/Users/nsmr2/AppData/Local/Temp/claude/C--Users-nsmr2--claude-spares-spare-N7tt8u/
cc0425f0-5907-4bc4-b41f-f529304b67cc/scratchpad/after_dbg.mp4: 601 fra ...
```

Tool call: Bash

```
# Fix double-tap demo and re-record after video
cd "/c/D/B.Tech/3rd year/SE" && SP="/c/Users/nsmr2/AppData/Local/Temp/claude/C--Users-nsmr2--claude-spares-
spare-N7tt8u/cc0425f0-5907-4bc4-b41f-f529304b67cc/scr ...
import sys
p = sys.argv[1]
s = open(p).read()
for old, new in [
    ('''                if d[1] == 0 and 2 < hy and 2 < hx < engine.grid_width - 3:
                    perp, rev = (0, -1), (-d[0], 0)''',
     '''                if d[1] == 0:
                    perp, rev = (0, -1) if hy > 2 else (0, 1), (-d[0], 0)'''),
    ('''                    print("no double tap: dir", d, "head", (hx, hy))
''', ''),
]:
    assert s.count(old) == 1, old
    s = s.replace(old, new)
open(p, "w").write(s)
EOF
python -B "$SP/record_demo.py" 45_snake Lab-4/videos/after.mp4 after 7 2>&1 | grep -v "Hello from\|pygame 2.6"
```

```
wrote Lab-4/videos/after.mp4: 601 frames (10.0s), marks={'rev': 125, 'tap': 170, 'dead': 321, 'hard': 446},
sounds=[(125, 'eat'), (286, 'eat'), (321, 'game_over ...
```

Tool call: Bash

```
# Extract preview frames and stream info from after video
cd "/c/D/B.Tech/3rd year/SE" && SP="/c/Users/nsmr2/AppData/Local/Temp/claude/C--Users-nsmr2--claude-spares-
```

```
spare-N7tt8u/cc0425f0-5907-4bc4-b41f-f529304b67cc/scr ...
```

```
  Duration: 00:00:10.02, start: 0.000000, bitrate: 140 kb/s
    Stream #0:0[0x1](und): Video: h264 (High) (avc1 / 0x31637661), yuv420p(progressive), 600x672, 104 kb/s,
  60 fps, 60 tbr, 15360 tbn (default)
    Stream #0:1[0x2](und): Audio: aac (LC) (mp4a / 0x6134706D), 44100 Hz, stereo, fltp, 22 kb/s (default)
  total 260
  drwxr-xr-x 1 nsmr2 197609      0 Oct 23 18:39 .
  drwxr-xr-x 1 nsmr2 197609      0 Oct 23 18:33 ..
  -rw-r--r-- 1 nsmr2 197609 176523 Oct 23 18:39 after.mp4
  -rw-r--r-- 1 nsmr2 197609  84850 Oct 23 18:33 before.mp4
```

Tool call: Read

```
{
 "file_path": "C:\\Users\\nsmr2\\AppData\\Local\\Temp\\claude\\C--Users-nsmr2--claude-spares-spare-N7tt8u\
\cc0425f0-5907-4bc4-b41f-f529304b67cc\\scratchpad\\aft ...
}
```

```
  [image]
```

Claude

Both videos look right. One cosmetic issue: the food square shows through the menu text, so I'm darkening the overlay, then updating the README.

Tool call: Bash

```
# Darken overlay and commit the polish
cd "/c/D/B.Tech/3rd year/SE/45_snake" && sed -i 's/^OVERLAY = (0, 0, 0, 180)$/OVERLAY = (0, 0, 0, 215)/' game/
game_engine.py && git diff && git commit -qam "Dar ...

Co-Authored-By: Claude Opus 5.5 <noreply@anthropic.com>" && git log --oneline -1
```

```
  warning: in the working copy of 'game/game_engine.py', LF will be replaced by CRLF the next time Git
  touches it
  diff --git a/game/game_engine.py b/game/game_engine.py
  index 6270517..f1e7b2f 100644
  --- a/game/game_engine.py
  +++ b/game/game_engine.py
  @@ -10,7 +10,7 @@ GREEN = (0, 200, 0)
   RED = (220, 60, 60)
   GRAY = (170, 170, 170)
   YELLOW = (255, 220, 90)
  -OVERLAY = (0, 0, 0, 180)
  +OVERLAY = (0, 0, 0, 215)

   # (name, snake moves per second); Medium is the original speed.
   DIFFICULTIES = [("Easy", 5), ("Medium", 8), ("Hard", 12)]
  warning: in the working copy of 'game/game_engine.py', LF will be replaced by CRLF the next time Git
  touches it
  227c452 Darken the game-over overlay so the menu stays readable
```

Tool call: Edit

```
file: C:\D\B.Tech\3rd year\SE\45_snake\README.md
--- old
---

## Expected Behavior
+++ new
---

## Changes Made

| Task | What changed | Where |
|------|--------------|-------|
| 1. Collision detection | Reversing into the body is ignored. Turns are checked against the last *queued*
turn, so two quick presses between moves (e.g. Up the ...
| Bug fix | The first food could spawn on top of the snake. Food now picks from free cells only. | `game/
food.py` |
| 2. Game Over | An in-game Game Over screen with the final score replaces the console `print`; the game waits
for input. | `game/game_engine.py` |
| 3. Replay | The Game Over screen offers Easy / Medium / Hard (5 / 8 / 12 moves per second) or Exit. The
current difficulty is shown top-right. | `game/game_en ...
| 4. Sound | A blip when food is eaten and a descending tune at Game Over, synthesised in code (no audio
files). The game stays silent instead of crashing if th ...

### Controls

| Key | Action |
|-----|--------|
| Arrow keys / `WASD` | Steer the snake |
| Up/Down (or `W`/`S`) + `Enter` | Pick an option on the Game Over screen |
| `1` / `2` / `3` | Replay on Easy / Medium / Hard |
| `Q` / `Esc` | Exit from the Game Over screen |

---
```

```
## Expected Behavior
```

Tool call: Bash

```
# Update README tree, add gitignore, untrack pycache
cd "/c/D/B.Tech/3rd year/SE/45_snake" && python - <<'EOF'
from pathlib import Path
p = Path("README.md")
s = p.read_text(encoding="utf-8")
old = """|   ├── snake.py
|   └── food.py"""
new = """|   ├── snake.py
|   ├── food.py
|   └── sound.py"""
assert s.count(old) == 1
p.write_text(s.replace(old, new), encoding="utf-8")
EOF
printf '__pycache__/\n*.pyc\n' > .gitignore && git rm -r -q --cached game/__pycache__ && git add .gitignore
README.md && git status --short
```

```
The file C:\D\B.Tech\3rd year\SE\45_snake\README.md has been updated successfully.

warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it
A  .gitignore
M  README.md
D  game/__pycache__/food.cpython-312.pyc
D  game/__pycache__/game_engine.cpython-312.pyc
D  game/__pycache__/snake.cpython-312.pyc
```

Tool call: Bash

```
# Commit README and gitignore update
cd "/c/D/B.Tech/3rd year/SE/45_snake" && git commit -q -F - <<'EOF'
Document the changes and controls; stop tracking __pycache__

Add a "Changes Made" table and a controls reference to the README, list
the new game/sound.py in the folder tree, and ignore Python bytecode
caches (the committed cpython-312 .pyc files are removed from git).

Co-Authored-By: Claude Opus 5.5 <noreply@anthropic.com>
EOF
git log --oneline && git status --short
```

```
6dace4c Document the changes and controls; stop tracking __pycache__
227c452 Darken the game-over overlay so the menu stays readable
42fe755 Task 4: Add sound feedback
af8034d Task 3: Add replay option with difficulty selection
bfb15f5 Task 2: Implement Game Over screen
3a63280 Fix food spawning on top of the snake
ceb9fc9 Task 1: Refine collision detection
114601f Initial Snake  starter
```

Claude

Re-recording the "after" video with the darker overlay and running all tests one more time.

Tool call: Bash

```
# Run all tests, re-record after video, check audio
cd "/c/D/B.Tech/3rd year/SE" && SP="/c/Users/nsmr2/AppData/Local/Temp/claude/C--Users-nsmr2--claude-spares-
spare-N7tt8u/cc0425f0-5907-4bc4-b41f-f529304b67cc/scr ...
```

```
task1 tests passed
food tests passed
task2 tests passed
task3 tests passed
task4 tests passed
wrote Lab-4/videos/after.mp4: 601 frames (10.0s), marks={'rev': 125, 'tap': 170, 'dead': 321, 'hard': 446},
sounds=[(125, 'eat'), (286, 'eat'), (321, 'game_over ...
[Parsed_volumedetect_0 @ 0000020112bedbc0] mean_volume: -21.1 dB
[Parsed_volumedetect_0 @ 0000020112bedbc0] max_volume: -5.8 dB
```

Tool call: Read

```
{
 "file_path": "C:\\Users\\nsmr2\\AppData\\Local\\Temp\\claude\\C--Users-nsmr2--claude-spares-spare-N7tt8u\
\cc0425f0-5907-4bc4-b41f-f529304b67cc\\scratchpad\\aft ...
}
```

```
[image]
```

Claude

Videos are done (after video has eat/game-over sounds mixed in). Now putting together the Lab-4 folder and the chat-history export.

Tool call: ToolSearch

```
{
 "query": "select:mcp__ccd_session_mgmt__export_transcript",
```

```
 "max_results": 1
}
```

Tool call: mcp__ccd_session_mgmt__export_transcript

```
{
 "session_id": "self"
}
```

```
   Exported this session's transcript "Lab 4 VibeCoding assignment" to C:\Users\nsmr2\Downloads\session-
   export-1792761158650.zip (1 MB, 1442818 bytes). The zip hol ...
```

Tool call: Bash

```
# Unzip the exported session transcript
SP="/c/Users/nsmr2/AppData/Local/Temp/claude/C--Users-nsmr2--claude-spares-spare-N7tt8u/cc0425f0-5907-4bc4-
b41f-f529304b67cc/scratchpad" && mkdir -p "$SP/export ...
```

```
   transcript.jsonl 2055840
   cc0425f0-5907-4bc4-b41f-f529304b67cc.jsonl 2055840
   cc0425f0-5907-4bc4-b41f-f529304b67cc/custom-title.json 45
   metadata.json 53489
   local-session-state.json 899
   ./cc0425f0-5907-4bc4-b41f-f529304b67cc/custom-title.json
   ./cc0425f0-5907-4bc4-b41f-f529304b67cc.jsonl
   ./local-session-state.json
   ./metadata.json
   ./transcript.jsonl
   Shell cwd was reset to C:\D\B.Tech\3rd year\SE
```

Tool call: Bash

```
# Inspect the transcript structure
SP="/c/Users/nsmr2/AppData/Local/Temp/claude/C--Users-nsmr2--claude-spares-spare-N7tt8u/cc0425f0-5907-4bc4-
b41f-f529304b67cc/scratchpad" && python -I - "$SP/exp ...
import json, sys, collections
rows = [json.loads(l) for l in open(sys.argv[1], encoding="utf-8")]
print(len(rows))
print(collections.Counter(r.get("type") for r in rows))
for r in rows[:6]:
    m = r.get("message", {})
    c = m.get("content")
    if isinstance(c, list):
        desc = [(b.get("type"), (b.get("text") or b.get("name") or str(b.get("content"))[:80])[:120]) for b in
c]
    else:
        desc = str(c)[:200]
    print(r.get("type"), m.get("role"), list(r.keys())[:12], desc)
EOF
```

```
   556
   Counter({'assistant': 116, 'attachment': 87, 'api-request-blob': 78, 'user': 67, 'api-request': 52, 'last-
   prompt': 29, 'relocated': 29, 'atis-latch': 29, 'custo ...
   queue-operation None ['type', 'operation', 'timestamp', 'sessionId', 'content'] None
   queue-operation None ['type', 'operation', 'timestamp', 'sessionId'] None
   user user ['parentUuid', 'isSidechain', 'promptId', 'type', 'message', 'uuid', 'timestamp',
   'permissionMode', 'origin', 'promptSource', 'turnOrigin', 'turnPosit ...
   https://github.com/SETAPESU26/45_snake
   this is what i have to do. do everything
   attachment None ['parentUuid', 'isSidechain', 'attachment', 'type', 'uuid', 'timestamp', 'rendered',
   'renderedRole', 'userType', 'entrypoint', 'cwd', 'sessionId ...
   attachment None ['parentUuid', 'isSidechain', 'attachment', 'type', 'uuid', 'timestamp', 'rendered',
   'renderedRole', 'userType', 'entrypoint', 'cwd', 'sessionId ...
   attachment None ['parentUuid', 'isSidechain', 'attachment', 'type', 'uuid', 'timestamp', 'rendered',
   'renderedRole', 'userType', 'entrypoint', 'cwd', 'sessionId ...
```

Tool call: Write

```
file: C:\Users\nsmr2\AppData\Local\Temp\claude\C--Users-nsmr2--claude-spares-spare-N7tt8u\cc0425f0-5907-4bc4-
b41f-f529304b67cc\scratchpad\transcript_to_pdf.py

"""Convert a Claude Code session export (transcript.jsonl) into a readable PDF.

usage: python transcript_to_pdf.py <transcript.jsonl> <out.pdf> <title>
"""
import html
import json
import re
import sys
from datetime import datetime

import pymupdf

src, out, title = sys.argv[1:4]
MAX_RESULT_LINES = 20
MAX_INPUT_LINES = 40

REMINDER = re.compile(r"<system-reminder>.*?</system-reminder>", re.S)
```

```
def clean(text):
    return REMINDER.sub("", text).strip()


def clip(text, max_lines):
    lines = text.splitlines()
    if len(lines) > max_lines:
        lines = lines[:max_lines] + [f"... ({len(lines) - max_lines} more lines)"]
    return "\n".join(line[:160] + (" ..." if len(line) > 160 else "") for line in lines)


def pre(text):
    return f"<pre>{html.escape(text)}</pre>"


def tool_input_text(name, inp):
    if not isinstance(inp, dict):
        return str(inp)
    if "command" in inp:
... (91 more lines)
```

File created successfully at: C:\Users\nsmr2\AppData\Local\Temp\claude\C--Users-nsmr2--claude-spares-spare-N7tt8u\cc0425f0-5907-4bc4-b41f-f529304b67cc\scratchpa ...

Tool call: Bash

```
# Convert transcript to PDF and render preview pages
SP="/c/Users/nsmr2/AppData/Local/Temp/claude/C--Users-nsmr2--claude-spares-spare-N7tt8u/cc0425f0-5907-4bc4-b41f-f529304b67cc/scratchpad" && python -I "$SP/trans ...
import pymupdf,sys; d=pymupdf.open(sys.argv[1]); p=d[0].get_pixmap(dpi=70); p.save(sys.argv[2]); p=d[3].get_pixmap(dpi=70); p.save(sys.argv[3])" "$SP/test_chat. ...
```

wrote C:/Users/nsmr2/AppData/Local/Temp/claude/C--Users-nsmr2--claude-spares-spare-N7tt8u/cc0425f0-5907-4bc4-b41f-f529304b67cc/scratchpad/test_chat.pdf: 17 page ...

Tool call: Read

```
{
 "file_path": "C:\\Users\\nsmr2\\AppData\\Local\\Temp\\claude\\C--Users-nsmr2--claude-spares-spare-N7tt8u\
\cc0425f0-5907-4bc4-b41f-f529304b67cc\\scratchpad\\cha ...
}
```

[image]

Tool call: Read

```
{
 "file_path": "C:\\Users\\nsmr2\\AppData\\Local\\Temp\\claude\\C--Users-nsmr2--claude-spares-spare-N7tt8u\
\cc0425f0-5907-4bc4-b41f-f529304b67cc\\scratchpad\\cha ...
}
```

[image]

Claude

The chat-history PDF converter works (17 pages so far). Copying the code into Lab-4 and writing its README, then I'll do the final export.

Tool call: Bash

```
# Skip task-list results in PDF; copy code into Lab-4
SP="/c/Users/nsmr2/AppData/Local/Temp/claude/C--Users-nsmr2--claude-spares-spare-N7tt8u/cc0425f0-5907-4bc4-b41f-f529304b67cc/scratchpad" && python -I - "$SP/tra ...
import sys
p = sys.argv[1]
s = open(p, encoding="utf-8").read()
for old, new in [
    ('''body = []
first_ts = None''', '''body = []
first_ts = None
skipped_ids = set()'''),
    ('''            elif block.get("type") == "tool_result":
                text''', '''            elif block.get("type") == "tool_result":
                if block.get("tool_use_id") in skipped_ids:
                    continue
                text'''),
    ('''                if name.startswith("mcp__ccd_turn__"):
                    continue''', '''                if name.startswith("mcp__ccd_turn__"):
                    skipped_ids.add(block.get("id"))
                    continue'''),
]:
    assert s.count(old) == 1, old
    s = s.replace(old, new)
open(p, "w", encoding="utf-8").write(s)
EOF
cd "/c/D/B.Tech/3rd year/SE" && rm -rf Lab-4/code && mkdir -p Lab-4/code && git -C 45_snake archive HEAD | tar -x -C Lab-4/code && find Lab-4 -type f | sort
```

Lab-4/code/.gitignore

```
Lab-4/code/README.md
Lab-4/code/game/food.py
Lab-4/code/game/game_engine.py
Lab-4/code/game/snake.py
Lab-4/code/game/sound.py
Lab-4/code/main.py
Lab-4/code/requirements.txt
Lab-4/videos/after.mp4
Lab-4/videos/before.mp4
```

Tool call: Bash

```
# List commit history oldest first
cd "/c/D/B.Tech/3rd year/SE/45_snake" && git log --reverse --format="%h  %s"
```

```
114601f  Initial Snake  starter
ceb9fc9  Task 1: Refine collision detection
3a63280  Fix food spawning on top of the snake
bfb15f5  Task 2: Implement Game Over screen
af8034d  Task 3: Add replay option with difficulty selection
42fe755  Task 4: Add sound feedback
227c452  Darken the game-over overlay so the menu stays readable
6dace4c  Document the changes and controls; stop tracking __pycache__
```

Tool call: Write

```
file: C:\D\B.Tech\3rd year\SE\Lab-4\README.md

# Lab 4: VibeCoding (Snake, repo `SETAPESU26/45_snake`)

I fixed the broken Snake game and added the four features from the repo README, working with an AI assistant
(Claude Code).

## Contents

| Path | Deliverable |
|------|-------------|
| `videos/before.mp4` | (a) 10 s of gameplay on the original code. Pressing the opposite direction key kills
the snake instantly, and the game then freezes with ...
| `videos/after.mp4` | (a) 10 s of gameplay after the changes. A reverse key press is ignored, a quick double
tap makes a safe U-turn, the Game Over screen show ...
| `code/` | (b) The updated game code |
| `PES1UG24AM170_Lab4_Chat_History.pdf` | (c) The exported chat history with the AI assistant |

The videos were captured frame by frame from the running game (`main.py`). The caption bar under the game
explains each moment and shows the key that was presse ...

## Run the game

```bash
cd code
pip install -r requirements.txt
python main.py
```

Commits (one per task)

```
114601f  Initial Snake  starter
ceb9fc9  Task 1: Refine collision detection
3a63280  Fix food spawning on top of the snake
bfb15f5  Task 2: Implement Game Over screen
af8034d  Task 3: Add replay option with difficulty selection
42fe755  Task 4: Add sound feedback
227c452  Darken the game-over overlay so the menu stays readable
6dace4c  Document the changes and controls; stop tracking __pycache__
```

`code/README.md` has a table of what changed for each task and a list of the controls.
```

```
File created successfully at: C:\D\B.Tech\3rd year\SE\Lab-4\README.md (file state is current in your
context — no need to Read it back)
```